How far does a learned surrogate of a flowsheet simulator transfer across ores? A closed-circuit, size-by-mineral engine, with cases and ore groups held out

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Technical report · v0.10.007, 31 October 2026 · Licence CC BY 4.0
CAOS open-research programme · mineral-processing simulator
Version DOI 10.5281/zenodo.23837152 · concept DOI (always latest) 10.5281/zenodo.23783845
Not peer reviewed. Source code, data and computational records:
github.com/fsantibanezleal/CAOS_OreFlow

Abstract

Learned surrogates of process simulators are often evaluated on random states from plants they have already seen. We compare that interpolation setting with leave-one-case-out and leave-one-ore-group-out evaluation for five models trained on a steady-state mineral-processing simulator. The simulator tracks mineral mass over 63 size classes, water, closed grinding circuits and unit-level balance closure; its limitations against published examples are reported alongside the comparisons. On a 3,072-state design over twelve authored cases, the multilayer perceptron gives the lowest interpolation recovery RMSE (3.38 percentage points), but has a 26.24-point mean RMSE when each case is held out. The Gaussian process has the lowest mean held-out-case RMSE (10.95 points), while the perceptron has the highest median case-fold R^2 (0.547). Under ore-group holdout, the perceptron's median recovery R^2 is 0.437 across seven groups, but varies from 0.795 for the five chalcopyrite cases to -10.2 for the two gold-in-pyrite cases. Log-upgrade transfer is poor for every model. Thus interpolation, case holdout and ore-group holdout answer different questions, and their results depend on both target and ore family. The records and tests bind every reported score to the bake. The cases are authored; these results do not establish performance on operating plants.

1 Introduction

A surrogate of a flowsheet simulator is attractive for the same reason as any surrogate: the simulator is slow and the surrogate is fast, so it can sit inside an optimizer, a sweep or an interactive tool. Whether it can stand in for the simulator depends on the question it is asked. Inside the envelope it was trained on, an interpolation score answers that question. A digital twin is asked something harder: what happens with a different ore, a harder or finer feed, another plant. That is transfer, and an interpolation score says nothing about it.

This work measures both on the same models, with a simulator whose answers are known exactly and can be reproduced. Its contributions are three:

1. an open flowsheet engine that represents streams by size and mineral, solves closed circuits, audits every balance from its output streams, is checked against published examples with every miss stated, and is reproduced by a line-by-line second implementation;

2. a test protocol that scores learned surrogates of it by interpolation, by leaving one case out and by leaving one ore group out over twelve authored plants in four circuit families, with interval coverage and an out-of-envelope guard;
3. the findings that holding out one plant is not a transfer test when other plants share its ore, that transfer failure is concentrated in the plants unlike the others, that the ranking of models changes between the protocols and with the statistic that summarizes them, and that the guard misses shifts of inputs its training features do not correlate with.

2 The engine

The record reproduces all 96 committed states within 1e-6 in the browser port and all 96 committed states close within 1e-9 at every unit in the Python engine. The port shares constants and is a translation check, not an independent validation of physical assumptions.

2.1 Representation

Every stream holds, for each mineral, a vector of mass flows (t/h) over one grid of 63 size classes on a fourth-root-of-two progression from 150 mm to a pan below 3.24 µm, and a water flow. Assays are computed from mineral masses and the minerals' element contents (from their formulas and the IUPAC atomic weights); no grade is stored, so a grade changes only when a mass does. Liberation is described per valuable mineral by size (King 1979): a liberated fraction, and composites whose content and host are declared.

2.2 Comminution

The crusher is the Whiten form (Duarte et al. 2021), with a strictly lower-triangular breakage matrix `B` and a classification function `C` set by the closed-side setting:

```
p = (I - C)(I - B C)^-1 f
```

The ball mill is an energy-specific population balance (Herbst and Fuerstenau 1980) in the form of the Moly-Cop tools: three perfect mixers of volume fractions 0.70, 0.15 and 0.15 share the breakage operator `D = (I - B) diag(S^E)`, so the mill's inverse transfer at an energy per pass `e` is a cubic polynomial in `D`, and the closed circuit with the cyclone returning a fraction `C_i` of each class solves

```
(T^-1(e) - diag(C)) p = f,        T^-1(e) = I + c1 e D + c2 e^2 D^2 + c3 e^3 D^3
```

per mineral. Two scalar unknowns, the energy per pass that meets the target P80 and the host cut that holds the design circulating load, are found by the Illinois method. The energy per pass is per tonne of mill feed, so the mill draws `e m`, with `m` the fresh feed plus the cyclone underflow (less what a gravity bleed takes from it); when that exceeds the installed power, the circuit runs at installed power, `e m(e) = P_inst` solved as a fixed point, with a coarser product, and says so. The reported grinding energy is that net energy per tonne of fresh feed, `e m / F`; the crushing energy is Bond's with the crushing work index, and the Bond requirement `E = 10 W_i (P^-1/2 - F^-1/2)` gives the operating work index and the efficiency ratio beside it, with Rittinger and Kick calibrated to agree with Bond at a reference reduction and never added to it (Bond 1952; GMG 2021).

2.3 Classification and separation

Hydrocyclones follow Plitt (1976): a Rosin-Rammler partition `y(d) = R_f + (1 - R_f)(1 - exp(-ln 2 (d/d50c)^m))` with a water bypass `R_f` from the water balance and a cut corrected for each mineral's density, so dense minerals classify as coarser particles and circulate. Plitt's sizing of the cyclone cluster is uncalibrated

and fails its published check (section 3), so the cluster's count and pressure are reported as an estimate, never as a result. Flotation banks are perfect mixers in series; the rate constant of each particle class follows from the bubble surface area flux `S_b = 6 J_g / D_32` (Gorain et al. 1997, 1999) and a size window (Trahar 1981), a composite floats at the liberated rate scaled by its exposed surface, `P_comp = P_V phi^(2/3)` with `phi` the valuable mineral's volume share, entrainment follows Savassi et al. (1998), and cleaner and recleaner tails return to the preceding bank. Each cell's residence is its pulp volume, less the gas hold-up, over the flow that leaves it as tail, which depends on what the cell floats, so it is solved per cell as a fixed point:

```
r_j = (k tau_j + ENT w_j) / (1 + k tau_j + ENT w_j)
tau_j = V (1 - eps_g) / Q_tail,j
R_N = 1 - prod_j (1 - r_j)
```

Gold follows the gravity-recoverable gold (GRG) model of Laplante, Woodcock and Noaparast as Vincent (1997) states it: the gold is a GRG component with its own size distribution, from a GRG test, plus a non-GRG component; the GRG's cyclone cut takes a density correction with an exponent of 1.0, fitted to measured GRG partitions where Stokes would give 0.5; the gravity unit on a bleed of the cyclone underflow, or of the mill discharge, recovers GRG by size at a per-pass recovery `R(d) = R_max (1 - exp(-(d/x_g)^2))`; and grinding moves GRG to non-GRG at a slowed breakage rate (Banisi et al.). Low-intensity magnetic drums capture liberated magnetite and composites by their magnetite content; desliming cyclones discard slimes ahead of flotation, with the water split following the solids sent down at a declared underflow density.

2.4 Balance closure and the contract

An independent recomputation calculates, from the output streams alone, the closure of every mineral, every species and the water at every unit and for the circuit, and of every size class at the units that do not break particles. The operating point is thirteen inputs declared once, with units, per-case bounds, steps, the families they apply to, a cross-field rule and bilingual messages; the exported declaration is interpreted identically by the engine's validator, the web service and the browser, so a state is accepted or rejected with the same code everywhere. A state the engine cannot solve, such as a classifier cut at which the mill cannot draw its installed power, is refused with a reason rather than answered.

3 Verification

Balances. All 96 committed states close within 1e-9 relative at every unit and for the circuit, as recomputed from the stored streams by the artifact checks.

The second implementation. A line-by-line TypeScript port runs the engine in the browser. Re-simulating every committed state from its own definition, it matches every metric, stream record, curve and kinetic record within 1e-6 relative, with the same flags. It is a translation by the same author that reads the same constant, atomic-weight and mineral files, so it catches translation errors and not modelling errors. An independent verification of the 0.08.001 implementation against its equations, one reviewer per part of the engine and every finding verified, found defects in both engines, among them the composite rate, the cell residence, the mill power basis and the desliming water split; each was fixed, the engine's in both languages, with a test that fails on the earlier code, and every number below is from the fixed engine.

Published examples. Each is labelled as a published example, not as plant data:

Example	Published	Engine
Moly-Cop BallSim base case (504 t/h, every published input)	net 7.71 kWh/t, gross 8.56; cut 183.3 µm; 6 cyclones at 53 kPa	net 7.30 kWh/t, 5.2% below, inside the 20% tolerance set before the first run; cut 188 µm; Plitt's uncalibrated sizing asks for 2 cyclones at 816 kPa

Example	Published	Engine
GMG worked examples of the operating work index	14.4 and 11.7 kWh/t	14.38 and 11.71 kWh/t
Laplante and Staunton gravity example, run like for like (80.4% GRG, the unit on 10 to 60% of the mill discharge)	GRG recovery 79.8 to 95.9%, GRG circulating load 2016 to 413%	GRG recovery 69.5 to 90.7%, GRG circulating load 696 to 84%: outside the 5-point and 35% tolerances
Zandrivierspoort magnetite (a different ore; the 45 µm product is a regrind)	64.9 and 69.0% Fe at 75 and 45 µm	63.8 and 67.8% Fe: a step of 4.0 points against 4.1, at levels 1.1 and 1.2 points below

The gravity example misses because the measured GRG of the case's ore (Snip) is very fine: 9 to 31% of the GRG, almost all finer than about 37 µm, leaves by the overflow; without the GRG finer than 25 µm the GRG recovery comes within 1.1 points from the 20% row on, and the circulating load stays 50 to 69% low. The magnetite example checks a direction and a proportion, not a level, because its ore is not the engine's case.

4 The test

4.1 The plants

Twelve authored cases in four circuit families: nine copper, nickel, zinc and refractory-gold rougher-cleaner flotation circuits (`rougher`), free-milling gold with a gravity bleed (`gravity_rougher`), fine magnetite with magnetic drums instead of flotation (`magnetic`), and phosphate with desliming ahead of flotation (`deslime_rougher`). Every parameter carries its unit and either a source or the label authored; none is fitted to a plant. Five of the twelve are copper sulphide plants on chalcopyrite, and the two gold plants carry most of their gold in pyrite (55%, beside electrum in the free-milling ore and arsenopyrite in the refractory one); the other five each have a payable and dominant carrier no other case shares.

4.2 The design, the features and the targets

For each case a scrambled Sobol sequence draws 256 states the contract accepts over the case's whole envelope and two ore factors (liberation size and floatability within their uncertainty half-widths), 3072 in all; the engine simulates every state. Each state is described by 22 physical features computed before simulation: the payable fraction, throughput per installed megawatt, the grind, circulating load, water, crusher setting and work index, the payable carriers' share-weighted liberation size, composite content, density, floatability and dose ratio, the rougher gas velocity, cells and volume per t/h, the gravity bleed, the desliming cut and four circuit descriptors. The case identity is never a feature. The targets are recovery (%), the log10 upgrade ratio (concentrate over head grade) and the total specific energy (kWh/t).

4.3 The models and the protocols

Ridge (alpha 1); a random forest (300 trees; Breiman 2001); histogram gradient boosting (300 iterations; Friedman 2001); a Gaussian process with a squared-exponential kernel with one length scale per feature and a noise term, on a seeded subsample of 500 training rows (Rasmussen and Williams 2006); and a perceptron of two hidden layers of 64 units, trained by full-batch AdamW with early stopping on a 15% validation split (Loshchilov and Hutter 2019). Features and targets are standardized on the training rows of each split. Because the Gaussian process sees fewer rows, the other models are also refitted on the same 500 rows; and because one training seed is one draw, the perceptron is retrained with four more seeds on every split.

Interpolation holds out 20% of the states inside every case (2460 training, 612 test rows). *Leave one case out* holds out each case in turn; the models that predict a case never saw any of its states, but they may have seen plants on the same ore. *Leave one ore group out* holds out every plant that shares the held-out plant's

payable and dominant carrier mineral: the five chalcopyrite plants together (1792 training, 1280 test rows), the two gold plants together (2560 training, 512 test rows), and each of the other five alone, where it is the same fold as leaving one case out; seven groups in all. An autoencoder (16, 6 and 16 units) trained on the features gives a guard: its threshold is the 99th percentile of the reconstruction errors of its validation rows; its false-alarm rate is measured on the 612 held-out in-envelope states, and its false-accept rate on 11 016 probes, each a held-out state pushed past the training range of one continuous feature, at six distances from 0.02 to 1.0 of the range, above the maximum and below the minimum.

5 Results

5.1 Interpolation and held-out cases answer different questions

Table 1. Recovery, RMSE in percentage points. The last column is the median R^2 over seven held-out ore groups.

Model	Interpolation RMSE	Interpolation R^2	Leave one case out: mean RMSE	worst case	median R^2	Leave one ore group out: median R^2
Ridge	9.46	0.617	13.68	23.05	-0.127	-0.774
Random forest	6.49	0.820	13.16	48.86	0.359	0.129
Histogram gradient boosting	4.60	0.909	12.66	56.61	0.512	0.287
Gaussian process	5.34	0.878	10.95	27.91	0.481	0.171
Perceptron	3.38	0.951	26.24	239.51	0.547	0.437

The perceptron has the lowest interpolation RMSE (3.38 points) and highest pooled interpolation R^2 (0.951), while its mean leave-one-case-out RMSE is highest (26.24 points) because the magnetite fold is difficult. The Gaussian process has the lowest mean leave-one-case-out error (10.95 points); the perceptron has the highest median fold R^2 (0.547). The five-seed record gives the MLP interpolation RMSE range of 3.00 to 3.54 points and mean held-out-case RMSE of 25.05 to 30.10 points. These are different summaries and support no single model ranking for every use.

5.2 Case-level errors identify where interpolation fails

Table 2. Recovery RMSE (points) on each held-out case.

Held-out case	Ridge	Random forest	Gradient boosting	Gaussian process	Perceptron
Soft copper porphyry	12.54	11.59	9.47	11.09	5.98
Hard copper porphyry	9.62	6.72	4.82	5.07	1.72
Free-milling gold with gravity	8.55	12.48	10.53	15.83	12.76
Fine magnetite concentration	21.01	48.86	56.61	27.91	239.51

Held-out case	Ridge	Random forest	Gradient boosting	Gaussian process	Perceptron
Nickel sulphide with serpentine slimes	13.47	8.49	6.80	6.66	6.61
Phosphate with clay slimes	15.85	10.04	10.05	10.83	8.93
Copper-molybdenum bulk flotation	9.38	6.12	4.03	6.33	1.91
Oxide copper: assumed sulphidized response	23.05	14.12	15.19	12.94	11.60
Zinc sulphide	10.87	11.88	10.18	8.49	7.26
Copper ore with clay	10.69	5.99	6.06	9.47	6.57
Low-grade copper at high throughput	9.68	6.25	4.48	6.41	2.23
Refractory gold in sulphides	19.44	15.41	13.77	10.32	9.86

The copper cases are not uniformly easy: holding out the soft porphyry costs the perceptron 6.0 points, while its error on the hard porphyry, molybdenum and low-grade copper cases is 1.7 to 2.2 points. Mixed high-clay copper is 6.6 points. The magnetite fold remains the largest perceptron error (239.5 points); the tree models also extrapolate poorly there. These model errors are measured against this authored simulator, not a plant assay.

5.3 Whole-ore-group transfer

When an ore group is held out, median recovery R^2 across seven groups ranges from -0.774 for ridge to 0.437 for the perceptron. The MLP scores 0.795 on the five chalcopyrite cases held out together and -10.2 on the two gold-in-pyrite cases; the group result is strongly dependent on ore family. Three of five models have a positive median recovery R^2 across groups, but no such aggregate removes the poor individual-group results. The release therefore reports per-case and per-group errors alongside pooled scores.

5.4 Energy and upgrade targets

The one-case-out median R^2 for total specific energy ranges from 0.652 (ridge) to 0.952 (gradient boosting); under ore-group holdout it ranges from 0.617 (ridge) to 0.907 (Gaussian process). Log-upgrade median R^2 is negative for every model under group holdout, so the recovery transfer result does not generalize to grade-upgrade prediction. The held-out protocols and target definitions remain separate in the record.

5.5 Interval coverage

The Gaussian process intervals are compared with their empirical coverage in the committed record. Coverage is target- and split-specific; the point estimates are not used as a substitute for uncertainty validation. The authored-plant split does not establish calibrated intervals on real plants.

5.6 The guard

The autoencoder guard is evaluated both on in-envelope states and on deliberately shifted probes. Its per-feature and per-distance acceptance rates are recorded in `learning.json`; a low reconstruction error is not treated as proof that an input shift is physically safe. The guard is a screening signal, not a process constraint or a substitute for the operating contract.

6 The method records

The engine produces, for each variant, records that do not depend on learning:

- **Kinetic lumping.** Five lumped models fitted to a virtual batch test of the rougher feed, projected to the bank under the same residence distribution and compared with the exact bank recovery by true flotation, over 88 flotation states: mean absolute projection errors of 1.27 points for the gamma model, 1.81 for Kelsall, 2.80 for Klimpel, 4.76 for the stretched exponential and 7.56 for first order (Polat and Chander 2000; Vinnett and Waters 2025). The first-order projection falls below the exact result at every nominal state: a batch curve that is a mixture of rates is not one exponential.
- **Constrained optimization.** A generalized pattern search with a progressive barrier from six starts (Torczon 1997; Audet and Dennis 2006, 2009) finds a point within the grade, power and water constraints for 94 of the 96 variants; the two exceptions are the magnetite circuit's harder ore and higher throughput, where the grind is the only decision and the mill is already at installed power. The learned lane screens the search step (Booker et al. 1999): candidates are ranked by the perceptron's prediction and reach the engine only where the guard accepts their state and the Gaussian process's 95% half-width on recovery is within 5 points, a novelty measure for the perceptron rather than an interval on what it ranks; the poll and the optimum stay engine results. The screen did not save engine work: over the 72 screened variants it used 26,367 engine evaluations versus 22,487 without screening, 17.25% more. It saved calls in none, used more in 68 variants and tied in four. Mean absolute recovery error averaged 0.66 percentage points across variants; across 5,811 proposals its mean was 0.71 points, median 0.48 and maximum 38.16. Both procedures had the same terminal outcome in 65 variants: 63 identical feasible decisions and two matching infeasible outcomes. Seven feasible pairs selected different decisions, with recovered-metal differences no greater than 0.002653 t/h (0.017%). Five favored the unscreened decision and two the screened decision.
- **Sensitivity.** Sobol indices of four uncertain ore properties at each nominal state (Saltelli et al. 2010; Herman and Usher 2017): floatability drives recovery in ten cases, liberation size drives concentrate grade in seven and the head grade in the other five, the work index drives grinding energy and the head grade drives recovered metal in all twelve.

7 Measured data, which calibrate nothing

The engine calibrates nothing and is calibrated by nothing. Measured data feed three lanes, and one of them also runs through the engine as inputs.

Particles. The HZDR constructed-case workbook (Pereira et al. 2021; RODARE 336, CC BY 4.0) keeps its training sheet (68 008 particles with A/B labels) and its test sheet (29 147 particles with constructed probabilities and the source authors' predictions, but no observed labels). A sparse logistic model per case and a shared network (4, 32, 32, 4) are scored against the constructed probabilities:

Case	Published reference	L1 logistic	Network
1	0.0257	0.0131	0.0138
2	0.0175	0.0157	0.0181
3	0.0207	0.0171	0.0143
4 (28 484 complete rows)	0.0371	0.1919	0.0226

The linear model beats the network in cases 1 and 2 and fails only in case 4, where the network does not.

Locked-cycle tests. 52 GeoMet locked-cycle copper recoveries from 29 drill holes of one deposit (Hoffimann et al. 2022; archived dataset 7051975, CC BY 4.0), predicted from five assays by the training mean, ridge, a random forest and a Gaussian process, under five folds of whole holes and three spatial zones. RMSE ranges from 5.09 to 5.51 points (whole holes) and 5.15 to 5.69 (zones). A paired bootstrap over complete holes (2000 resamples) shows how little separates them. On the five fixed hole folds ridge beats the training mean by 0.42 points (95% interval 0.02 to 0.82), but that partition sits at the 98.5th percentile of 200 random hole partitions; averaged over them the gain is 0.17 points (-0.39 to 0.71) and under leave one hole out 0.22 (-0.36 to 0.79), each interval widened for the six model pairs, and under spatial zones no model separates either. The source has no grind, reagent or residence information, so it cannot calibrate the engine's controls.

The engine on the same samples. The 52 samples also run through the engine as inputs, each on its own assays in the soft copper porphyry's circuit at that case's operating point; nothing in the engine is fitted to them. The work index comes from the nearest comminution sample in the hole (42 samples) or the deposit median (10), and spans 15.2 to 26.1 kWh/t against the case's 11.0. Most of the copper is bornite and chalcocite, allocated by a sulphur-limited normative mineralogy, one choice in a family of allocations; the two minerals float at declared ratios to chalcopyrite (0.8, and chalcocite at 1.5 times bornite, a choice the GeoMet data do not bound). Every sample runs the mill at installed power, and the engine's recovery falls short of the locked-cycle test by 22.6 points on average (RMSE 24.1 points; 2.9 to 41.1 points below). That gap is mostly the host circuit's size: the samples are harder than the circuit's design ore, so at 720 t/h the product is coarse. With the circuit sized for the case's 150 µm target the gap is +0.2 points, and it runs from +10.0 to -20.8 points as the assumed laboratory grind goes from 75 to 300 µm, which the open data do not give. At each sample's own target-grind throughput it is +4.0 points, of which 3.8 come from the longer flotation residence at the lower throughput rather than from the grind. At every assumed grind the engine carries no information about which sample recovers more (Pearson r between 0.03 and 0.06). It is a comparison, not a calibration: the circuit, the breakage, the liberation and the flotation are authored for another ore, and a locked-cycle test is not a plant.

Plant hours. One iron-ore plant's reverse flotation record (Kaggle dataset 6294, CC0; 737,453 rows) gives hourly laboratory silica beside 21 feed, reagent, pulp and column sensors. Dropping the 310 hours whose silica label was interpolated leaves 3,701 pairs of consecutive hours. Scored on three future windows after at least 24 hours of embargo, the previous assay alone forecasts the next hour's silica with a mean absolute error of 0.464 points, and the best sensor-only model, ridge, is 0.001 points below the training mean's 0.766. In 461 of those pairs the laboratory value is one carried over unchanged for three or more consecutive hours; without them the previous assay's error is 0.510 points, still the lowest, and ridge is 0.014 points above the training mean. The engine has no reverse cationic flotation family, so these hours are shown and never simulated.

8 Discussion and limitations

The test bounds transfer between authored plants. A thirteenth authored plant is not a real one: the engine is a steady-state model with authored parameters, no froth stability, no dynamics and no prices, and its twelve plants share its structural assumptions. That is a feature of the design, since the teacher's answer is exact and reproducible, and a limit of the conclusion, since a real plant can depart from all twelve at once.

Four findings should carry over to any surrogate of a flowsheet model. First, interpolation and transfer are different questions, and the model that answers the first best can answer the second worst. Second, holding out one plant tests transfer only to the extent that no training plant shares its ore: five copper plants on one mineral each look transferable alone and transfer far worse when held out together, so the held-out unit has

to be the ore, not the plant. Third, transfer error is an average over held-out plants that hides where it fails; the plants unlike the others dominate it, and reporting the worst held-out case matters as much as the mean. Fourth, a reconstruction guard misses shifts in inputs that do not move with the others, so its false-accept rate must be reported by input and by distance.

A calibrated study would replace the authored targets by measured metallurgy across operating periods, calibrate the engine on some periods and hold out an entire period and an ore family, and score both point error and whether the surrogate preserves the ranking of operating decisions.

9 Reproducibility

The repository is https://github.com/fsantibanezleal/CAOS_OreFlow (MIT). The engine, the bake, the records, the browser port, the service and the documentation are versioned together; the source, records and computational environment are versioned with the software release. `./scripts/setup.ps1` builds the environments and `./scripts/precompute.ps1` regenerates every record. On a workstation with 32 logical cores and an RTX 4070 Laptop GPU, the GPU-accelerated final MLP fit is a computational component of a simulator study over authored cases; it is not plant deployment or evidence of plant-calibrated performance. `./scripts/smoke.ps1` runs the checks: the guards, the Python and frontend suites and the build, with the counts of each release in `docs/release-verification.md`. The workbench at https://oreflow.ml.fasl-work.com runs the engine in the browser on any state of any case.

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